INCH-POUND
A-A-50198B
11 January 2017
SUPERSEDING
A-A-50198A
21 June, 2000

COMMERCIAL ITEM DESCRIPTION

THREAD, GIMP, COTTON, BUTTONHOLE

The General Services Administration has authorized the use of this commercial item description in preference to Federal Specification V-T-280.

1. SCOPE. This document covers thread intended for use as the reinforcing cord for eyelet-end buttonholes.

2. CLASSIFICATION. The following types of thread are as specified.

2.1 Types.

Type I - Soft Finish
Type II - Glazed Finish

3. SALIENT CHARACTERISTICS.

3.1 General description. The thread shall be made from cotton and shall have a soft finish for Type I or a glazed finish for Type II. Type I thread shall have no consistency or luster not inherent to the cotton fiber. Type II thread shall be polished and lustrous, with a smooth surface having no protruding fibers. Unless otherwise specified, the direction of the final twist shall be "S".

3.2 Physical requirements. The finished thread shall conform to the applicable physical requirements as specified in Tables I and II, when tested as specified in Table II.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to: DLA Troop Support Standardization Team, 700 Robbins Avenue, Philadelphia, PA 19111-5096. Since contact information can change, you may want to verify the currency of this address information using Acquisition Streamlining and Standardization Information System (ASSIST) online database https://assist.dla.mil.

AMSC N/A FSC 8310

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:40Z
Check the source to verify that this is the current version before use.

TABLE I. Physical requirements.

Finish of Thread	Tex (sizes)	Ply (min.)	Breaking strength, lbs. (min.)
Type I - Soft	180	3	6.5
	210	3	6.5
	270	4	9.0
	350	5	11.3
Type II - Glazed	180	3	6.8
	210	3	6.8
	300	4	9.4
	350	5	11.8

3.2.1 Physical requirements and test methods. The thread shall conform to the physical requirements specified in Tables I and II, when tested in accordance with Table II.

TABLE II. Physical requirements and test methods.

Characteristic	Requirement	Test method
Fiber identification	3.1	ASTM D276 or AATCC 20 1/
Direction of twist	3.1	ASTM D204
Tex (Size)	Table I	ASTM D204
Ply	Table I	Visual
Breaking strength, pounds	Table I	ASTM D204 2/, 3/
Colorfastness (all Types):		
Laundering (after 3 cycles)	3-4	AATCC 61 Test 3A 4/, 6/
Wet drycleaning	3-4	AATCC 132 4/, 5/

1/ In case of dispute, the ASTM method prevails
2/ Testing speed shall be 12 (± 0.5) in/min., and a 10-inch gauge length shall be used
3/ Except that five (5) determinations shall be made per sample unit.
4/ AATCC Evaluation Procedure 1, Gray Scale for Color Change
5/ AATCC Evaluation Procedure 2, Gray Scale for Staining
6/ The test specimens shall be dried after each laundering cycle. The color transfer cloth evaluation shall not apply. No appreciable change in shade or loss of color of the tested specimen shall be visible when compared to the untested thread retained.

3.3 Knots. The thread shall have an average of not more than one (1) thread knot or slice per 1,000 yards.

3.4 Color. The color shall be as specified in the applicable end item document or in the contract. The dyed thread shall conform to the applicable approved shade standard (see 3.6).

3.5 Labile Sulfur. Dyes and compounds containing elementary sulfur capable of oxidation to sulfuric acid shall not be used.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:40Z
Check the source to verify that this is the current version before use.

3.5.1 Presence of labile sulfur test. In the determination of presence of labile sulfur in textile materials with lead acetate, two (2) 1.50 (± 0.01) gram samples from each material submitted for evaluation shall be tested. Each of the two (2) samples shall be cut into very small pieces and placed into separate test tubes. The samples shall be submersed in a stannous chloride solution that contains 100 grams of stannous chloride crystals ACS in 100 milliliters of hydrochloric acid ACS (35 percent concentration) and 50 milliliters of distilled water. A filter paper wet out with a 5.0 percent lead acetate solution shall be placed over top of the test tube. The lead acetate solution contains 5.0 grams of lead acetate CP reagent grade and enough distilled water to make up a 100 milliliter solution; if the solution is not clear, add a few drops (one at a time) of glacial acetic acid until the solution is clear. The test tube containing the textile sample, stannous chloride and wet filter paper shall be heated over a low flame until the solution is boiling. The solution should not be heated for more than 15 seconds. A brown to black stain on the filter paper should be evaluated as follows:

Free	-	The filter paper shows no discoloration or staining of any kind.
Slight	-	The filter paper shows a light tan to light brown discoloration stain.
Moderate	-	The filter paper shows a dark brown discoloration stain.
Severe	-	The filter paper shows a black color stain.

3.6 Visual shade matching. The color and appearance of the finished thread shall match the standard sample when viewed using AATCC Evaluation Procedure 9, Option A, with sources stimulating artificial daylight D75 illuminant with a color temperature of 7500K (± 200), with illumination of 100 (± 20) foot candles, and shall be a good match to the standard sample under incandescent A illuminant with a color temperature of 2856K (± 200).

3.7 Finishing materials. The finished thread shall have no chemical finishes or treatments other than those commonly used (e.g. water-repellent, etc.) on commercial threads or as specified in the contract which have been demonstrated to have no deleterious effects on the fiber, including effects of prolonged storage. No finish or treatment shall be applied for the purpose of increasing breaking strength. There shall be no noticeable wicking of the treatment on the thread to adjacent material when sewn.

3.8 Toxicity. The finished thread shall not present a health hazard and shall show compatibility with prolonged, direct skin contact when tested as specified in 3.8.1. Chemicals recognized by the Environmental Protection Agency (EPA) as human carcinogens shall not be used.

3.8.1 Toxicity test. When required (see 7.6), an acute dermal irritation study and a skin sensitization study shall be conducted on laboratory animals. When the results of the studies indicate the thread is not a sensitizer or irritant, a Repeat Insult Patch Test shall be performed in accordance with the Modified Draize Procedure (see 7.2.4). If the toxicity requirement (see 3.8) can be demonstrated with historical use data, toxicity testing may not be required (see 7.6).

3.9 Put-up. Unless otherwise specified, the thread shall be put-up on holders such as commercial spools, cones, or tubes as specified in the contract. The thread shall be wound around the specified holder in one (1) continuous piece, so that each turn and layer is free of

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:40Z
Check the source to verify that this is the current version before use.

entanglement. The outside ending of the thread shall be secured to prevent unwinding, loosening, or slippage during handling, shipping, or storage.

3.10 Labeling. Each holder shall have a label, adhered securely so as to remain in place and be clearly legible until all thread has been removed. The label shall be printed and include information related to length in yards or weight of cone, direction of twist, color, Tex size, name of thread manufacturer, and nomenclature specifying fiber and construction Type. Type, stock number, date of manufacture (month and year) shall be on the label.

3.11 Workmanship. The finished thread shall conform to the quality of product established by this document. The occurrence of defects shall not exceed the contractor's own quality assurance standards and the quality assurance standards defined by the technical data in the bid package.

4. REGULATORY REQUIREMENTS. Unless otherwise specified the offer/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

5. PRODUCT CONFORMANCE PROVISIONS.

5.1 Product Conformance. The products provided shall meet the salient characteristics of this Commercial Item Description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance.

5.2 Visual examination. Thread shall be examined for the defects listed in Table III below.

TABLE III. Visual examination defects.

Knots: The thread shall average not more than one (1) full thread knot or splice per 1,000 yards for all Tex sizes.
Color: Not as specified.
Labels: Label missing, incorrect, or illegible. Required information missing from the label.
Packaging: Not packaged in accordance with the contract or purchase order.

5.3 Acceptance criteria. Acceptance criteria shall be as specified in the contract or purchase order (see 7.6).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:40Z
Check the source to verify that this is the current version before use.

6. PACKAGING.

6.1 Packaging. Preservation, packing, and marking shall be specified in the contract or order (see 7.6).

7. NOTES.

7.1 Sources of Government documents.

7.1.1 EPA Pollutants/toxics subtopics are available online at https://www.epa.gov/air-research/research-health-and-environmental.

7.1.2 Federal Acquisition Regulations are available online http://acquisition.gov/far/index.html.

7.2 Sources for Non-Government documents.

7.2.1 AATCC Test Methods are available online at http://www.aatcc.org.

7.2.2 ASTM Standards are available online at http://www.astm.org.

7.2.4 Repeat Insult Patch Test. Modified Draize Procedure - Principles and Methods of Toxicology, A Wallace Hayes (editor).

(Copies of this document are available online at http://www.taylorandfrancis.com/ or https://www.crcpress.com).

7.3 Applications. The thread is intended for machine sewing of clothing for buttonholes, tentage, equipage, footwear, and parachute applications.

7.4 Standard samples. For access to standard shade samples of thread, address the contracting activity issuing the invitation for bids or request for proposal.

7.5 Certificate of compliance. The contracting activity may select to accept a certificate of compliance for stated requirement.

7.6 Ordering data. The contract or purchase order should specify the following:

a. Title, number, and date of this Commercial Item Description.
b. Type of thread required (see 2.1).
c. Color required (see 3.4).
d. When toxicity testing is required (see 3.8).
e. Put-up required if other than specified (see 3.9).
f. Product conformance provisions (see 5.1).
g. Acceptance criteria provisions (5.3).
h. Packaging requirements (see 6.1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:40Z
Check the source to verify that this is the current version before use.

7.7 Key words.
Clothing
Eyelet-end
Machine sewing

Custodians:
Army – GL
Navy – NU
Air Force – 11

Preparing activity:
DLA–CT

(Project 8310-2016-004)

Review activities:
Army – MD
Navy – MC

CIVIL AGENCY COORDINATING ACTIVITY:

Civil agencies:
GSA – FAS

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:40Z
Check the source to verify that this is the current version before use.